

PRACTICAL COURSE WORKBOOK

Building AI Agents & LLM Apps

Build a responsible AI workflow for a real work task

LEVEL	Advanced
GUIDED TIME	5 hours across 12 lessons
PROJECT	a documented prompt, verification and revision workflow
SKILLS	LangChain, AutoGPT, Agents, Critical evaluation, Documentation
EDITORIAL STATUS	Structured draft v0.9

WHAT YOU WILL BE ABLE TO DO

- Use LangChain appropriately in a realistic, bounded task.
- Use AutoGPT appropriately in a realistic, bounded task.
- Use Agents appropriately in a realistic, bounded task.

WHO THIS IS FOR

People who want dependable, responsible AI-assisted workflows.

BEFORE YOU BEGIN

- Basic computer and web skills

This open workbook is a structured draft undergoing course-specific editorial review. It does not provide certification.

COURSE MAP

From foundation to finished work

Building AI Agents & LLM Apps is a structured, practical course covering LangChain, AutoGPT, Agents. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 LangChain: Foundations	1. Capabilities with LangChain 2. Limitations with AutoGPT 3. Responsible use with Agents
02 AutoGPT: Working methods	1. Clear instructions with LangChain 2. Context design with AutoGPT 3. Verification with Agents
03 Agents: Applied workflows	1. Research with LangChain 2. Creation with AutoGPT 3. Automation with Agents
04 LangChain: Quality and review	1. Evaluation with LangChain 2. Privacy with AutoGPT 3. Repeatable practice with Agents

HOW TO STUDY EACH LESSON

- 1 Read the objective and identify the decision it supports.
- 2 Reproduce the worked example with the stated input.
- 3 Test one normal case and one boundary or failure case.
- 4 Complete the knowledge check without guessing.
- 5 Save a short evidence note: result, limitation and next action.

CAPSTONE PROJECT

Produce a reviewable Building AI Agents & LLM Apps project using LangChain, AutoGPT, Agents.

DELIVERABLE	A working Building AI Agents & LLM Apps artefact plus an evidence-based self-review.
TOOLS	A suitable LangChain environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using LangChain.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

Evidence log

INPUT	Original brief, data, requirements or test material.
DECISIONS	Chosen method and one reasonable alternative.
TESTS	Normal, boundary and failure results.
LIMITATION	What the result does not establish.
REVISION	One evidence-led improvement.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

Authoritative references

AI Risk Management Framework 1.0

NIST - reviewed 2026-08-21

<https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-ai-rmf-10>

AI Principles

OECD - reviewed 2026-08-21

<https://www.oecd.org/en/topics/ai-principles.html>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the LangChain scope.